A Better Signal Is Not a Better Router: Pre-Generation Escalation Decisions in Sub-Billion VLMs

Theophile Avenel
github.com/thekester/gwel

Abstract

Vision–language cascades answer from a downsampled image and escalate to full resolution when the model appears uncertain. Every published escalation policy learns this decision by reinforcement learning on 7B-class models, and reads it from the model's output distribution after generation. We ask what a 500M-parameter model already knows, before it writes a token.

Output confidence answers the wrong question. It predicts whether the cheap pass will be *correct* (AUROC 0.758) and barely predicts whether escalating would *recover* a failed query (0.617), which is the quantity that actually spends budget. The distinction is invisible for text LLMs, where the intervention is extended reasoning over an unchanged input, and material here, where it adds information the model has not seen.

A linear probe on pre-generation activations reads that quantity better (0.760). **Then it fails an audit we ran on ourselves, and that failure is the paper's main result.** Escalation value is dataset-determined, DocVQA repairs 45% of its queries and VQAv2 5%, and image size tracks the same split, so raw image size, a feature costing no forward pass, reaches 0.748. Trained and tested inside one dataset the probe reaches 0.572 where output entropy, fitted on nothing, reaches 0.663; it improves with data and never closes the gap, at any depth, with any of four probe families including non-linear ones. **The probe is largely a domain detector**: where it and entropy disagree, the disagreement predicts which dataset a query came from at AUROC 0.738. On a benchmark mixture it routes well; on the homogeneous workload a deployment serves, the signal it beat is the better one.

The method survives the signal. Calibrating a score to the *expected signed gain* and thresholding at the break-even implied by measured latencies gives a policy with no rate to tune, cost-optimal among policies that see only the score (Proposition 1). Run on output entropy inside each domain, one unchanged operator preference yields escalation rates from 1% to 61% that track what escalation is worth there ($r = 0.974$), and dominates a fixed tuned rate. Calibrating the weak pass's error probability instead, which is the standard cascade recipe and coincides with ours exactly when escalation cannot hurt, spends 59% more compute for the same answers.

How far to escalate matters more than whether. On 1200 document pages with rungs chosen by measuring the model's patch-grid buckets, 93% of what escalation repairs is repaired below the top rung, and the top step gains -0.002 $[-0.026, +0.022]$ for 156 ms. That ceiling holds across an 8.6× parameter range and two distinct vision encoders, at the same pixel resolution but different token counts, so it is a property of the pages rather than of the model, and can be measured once per corpus.

The same signal does not say *where* to look. A localizer pooling visual-token states over candidate crops beats random cell choice at the finest grid we test, at every depth, by 0.9 accuracy points: 6% of the gap a perfect localizer would close. That gap is worth having, since at matched token budget the best crop beats the dearest resolution setting at 2.5× fewer tokens.

Every comparison here is a cost comparison, so we measure the cost model rather than assume it, and four standard proxies mispredict: the vision encoder is 43–50% of the visual cost, capping post-encoder pruning at half; decode latency tracks depth rather than parameters; a flat per-configuration price understates escalation by 16%; and on 56% of the pilot the "full-resolution" configuration is not a distinct configuration at all. It was that third correction, not a literature review, that exposed the domain confound. Fifty-six claims, including the negative ones and the limits of the positive ones, are re-derived from the data by a harness that fails loudly when any stops holding, with paired comparisons corrected for family-wise error.

1 Introduction

A vision–language model asked "what does the sign say?" can answer from a thumbnail or from the full-resolution image. The thumbnail costs a fraction as much and is often sufficient. Deciding *per query* which to use is the premise of adaptive visual perception, and the premise is sound: on our pilot, 34% of queries are answered correctly at 384 px and would gain nothing from more pixels, while 41% fail at both resolutions and would gain nothing either. Escalation changes the answer on 21%.

Published systems make this decision by learning it and by reading it after generating: VisionThink [1], AdaptVision [2] and AwaRes [3] all train an escalation policy with reinforcement learning on 7B-class models, supervised by judge models of their own.

Below one billion parameters none of that is available. Curating twenty thousand samples and running GRPO is not an option for a model that exists because it fits in a gigabyte of GPU memory [9]. Our question is therefore narrow: *what does a 500M VLM already know about its own need for more pixels, and how cheaply can that be* *read?*

Fig. effig:method is where we end up: a single cheap pass, a score, and a choice among resolution rungs priced by measured latency. The short answer changed while we were writing it. On a benchmark mixture the model appears to know more than its output reveals; on a single workload most of that apparent knowledge turns out to be a recognition of which benchmark the query came from. What survives is narrower and, we think, more useful: reading the right quantity matters more than reading it well, deciding *how far* to escalate matters more than deciding *whether*, and what the model knows is whether to look again rather than where.

Contributions.

1. **An audit that overturns our own headline.** Pricing escalation per example rather than per configuration reveals the probe escalating 1.19–1.33× larger images than entropy. Following that thread: free image size scores 0.748 against the probe's 0.761, and inside a single dataset the probe reaches 0.572 where output entropy, fitted on nothing, reaches 0.663, at any depth and with any of four probe families. The pre-generation advantage is a between-domain effect (§6, §6.1).
2. **Two questions, not one.** Confidence predicts whether the cheap pass is correct; it barely predicts whether escalation would fix it. The distinction is invisible for text LLMs, where the intervention is "think longer" over the same input, and material for VLMs, where it adds information the model has not seen (§3.3).
3. **A decision rule instead of a tuned rate**, and a third way to get the target wrong. Calibrating the expected gain and thresholding at the cost-implied break-even removes the rate parameter; calibrating the weak model's error probability instead, the standard cascade recipe, over-spends by 59% relative for no accuracy, because it assumes escalation cannot make things worse (§7). The rule is a property of the method, not of the probe: run on output entropy inside each domain, one unchanged operator preference yields escalation rates from 1% to 61% tracking what escalation is worth there ($r = 0.974$), and dominates a fixed tuned rate (§7.1).
4. **How far to escalate is the decision the field collapses.** Of the queries escalation repairs, 77% are repaired by an intermediate rung, and that rung is twice as efficient per millisecond as the one above it. The rule generalises to a ladder without modification and saves 35 ms at no accuracy cost on the one dataset where a top rung exists, which is also how we discovered that for 56% of the pilot the "full-resolution" configuration is not a distinct configuration at all (§7.2).
5. **A pre-generation probe, and the target it must be trained on.** AUROC 0.760 against 0.617, saturating at layer 6 of 32 on the mixture, though §6.1 shows that depth does not transfer. Trained on the conditional target it beats entropy on ranking and loses to it as a policy by 8.5 points; trained on the joint target it holds 90% of the Pareto front and is cheaper at matched accuracy in every one of 200 resplits (§5, §5.1).
6. **Two ways a better signal fails a policy**, both measured: a target the policy does not face, and a signal the policy cannot afford. An entropy–probe ensemble ranks best of all and is dominated as a policy, because reading it requires the pass it was meant to shorten (§5.1).
7. **A near-negative on localization, stated as a magnitude.** Pooling visual-token states per candidate crop beats random cell choice at the finest grid we test, at every depth, by 0.9 accuracy points: 6% of the gap a perfect localizer would close (§8).
8. **A measured cost model.** Every comparison here is a cost comparison, so we measure rather than assume: the vision encoder is 43–50% of the visual cost across an 8.6× parameter range, which caps post-encoder pruning at half, and decode latency follows depth rather than parameter count (§3.2).
9. **The untrained escalation mechanism is weak.** Asking the model whether it needs more resolution, the mechanism published systems train, escalates as often as uncertainty-based routing and selects the wrong queries, losing to the probe by 0.070 $[0.020, 0.120]$ accuracy at 25.8 ms higher cost (§5.1).
10. **An ablation read at fixed budget, not fixed preference.** Held at a fixed operator preference, every ablation appears to improve accuracy, because every removal escalates more. Read at a fixed latency budget, the ladder is worth 37 accuracy points and per-example pricing 0.003 (§A).
11. **Abstention priced on both axes.** Declining a query is the weaker lever for holding risk down, escalation reaches coverage abstention cannot, and is nearly worthless as a budget action, because a gain-calibrated rule already escalates unanswerable queries below their base rate, leaving a perfect gate only 4.2% of latency to recover (§D).

2 Related work

Adaptive visual acquisition. The three systems closest to this work all learn when to look closer. VisionThink [1] trains a special token requesting the full-resolution image with GRPO and an LLM-as-judge reward, plus a penalty calibrated to stop the policy collapsing into

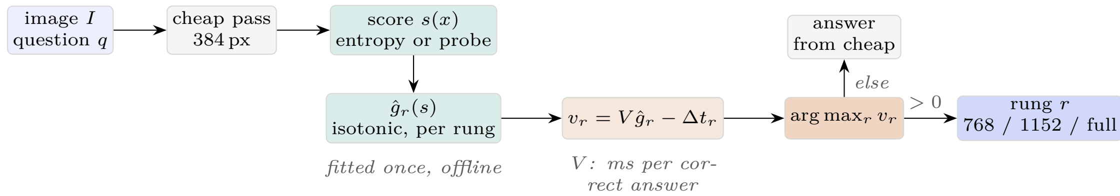

Figure 1: The serving path. Everything up to the score is a single cheap pass; the calibrators are fitted once offline and the operator supplies one number, V. Which rung is chosen, and whether any is, falls out of the measured latencies rather than a tuned rate (Algorithm 2).

always escalating. AdaptVision [2] replaces the binary decision with a bounding-box tool call and introduces Decoupled Turn Policy Optimization to separate the credit assigned to tool tokens from that assigned to answer tokens. AwaRes [3] makes the coupled decision of *whether* and *where* jointly, with cold-start supervised fine-tuning followed by multi-turn GRPO, supervised automatically by a LLaMA-3.3-70B judge and an oracle grounding model. All three evaluate at 7B scale, all three learn the decision rather than reading it, and all three condition on the output of a completed pass. We differ on each count, and §5.1 measures what the first of those choices is worth by implementing their mechanism without its training.

Visual token compression. A parallel line reduces cost by discarding tokens after the vision encoder has run: FastV and SparseVLM prune on attention scores, and GlimpsePrune [4] learns a data-driven pruning metric from a single-pass "glimpse", removing 92.6% of visual tokens at parity. These methods and ours act on different halves of the same cost, which §3.2 quantifies: post-encoder pruning cannot recover the encoder's share, and on the three models we measure that share is 43–50%.

Routing from internal states. In the text setting, several groups have shown that a model's activations predict its own success before it generates. Lugoloobi et al. [5] train linear probes on pre-generation activations and route across a model pool at 40–70% lower cost; Moreno Cencerrado et al. [6] isolate a linear "in-advance correctness direction" that saturates at intermediate layers; NVIDIA's LLM Router [7] predicts a target model's correctness from a *different* model's prefill activations, closing 45.6% of its oracle gap. Every probe in this literature predicts whether the model will succeed. None predicts whether an intervention would change the outcome, because for text LLMs the intervention is extended reasoning over an unchanged input and the two questions nearly coincide. For a VLM they do not, and §3.3 shows they come apart empirically.

Where this sits. Reproducing GRPO training at 7B is not affordable here, so we do not claim a numeric comparison against the systems below; what can be stated

method	reads	when	trained by
VisionThink [1]	special token	during	GRPO
AdaptVision [2]	tool call	during	DTPO
AwaRes [3]	policy output	during	SFT+GRPO
GlimpsePrune [4]	glimpse token	$\frac{2}{3}L$	a head
entropy baseline	output scores	after	n/a
ours	residual stream	layer ℓ	n/a

Table 1: What each method reads and what it costs to train. "During" means the decision is a generated token, available only once decoding starts. The three trained policies also need a supervisor: an LLM judge, GPT-4o, and a LLaMA-3.3-70B judge with an oracle grounding model respectively. GlimpsePrune prunes tokens rather than choosing resolution, and is listed because it is the one method that also reads an intermediate layer.

precisely is what each one reads, when it decides, and what supervision it needs (Table 1).

The distinction that matters for cost is the third column. Three of the four learn a decision the model *emits*, which requires generation to begin; ours and GlimpsePrune's read a state the forward pass has already computed. We note that GlimpsePrune measures its own overhead end to end and reports prefill at 69.1% of baseline, so the discipline of charging for the decision is not unique to this work, only unusual among the resolution-choosing methods.

Calibrated cascade routing. A separate line takes the routing score as given and asks how the threshold should be set. Kotte [11] argues the threshold should not be tuned at all: calibrate the uncertainty signal to a probability by isotonic regression, and threshold policies on that probability are cost-optimal under three stated assumptions. Ruan et al. [12] set gates by an exact lower bound on the survival rate of successful cases, so the policy carries a recall guarantee and cannot degenerate into never escalating, and Tayebati et al. [13] obtain distribution-free coverage for a three-way answer/hedge/abstain split. §7 adopts the first and shows why its central assumption fails for a visual cascade.

Attacks on the allocation mechanism. Liu et al. [15] learn a universal trigger, confined to an image border, that lowers a weak model's confidence and so forces deferral to the expensive model: the provider pays and accuracy metrics never register it. Their attack is metric-agnostic because it flattens the output distribution itself, degrading perplexity, token-margin and verbalised confidence alike; Sun et al. [16] report the same failure mode for text cascades. Both target signals read *after* generation, which is where every visual escalation method we have surveyed reads.

Cost measurement. Zhan et al. [8] profile on-device VLM energy and find average power to be a model fingerprint, so energy reduces to time; their headline that decode dominates 86–97% of it is measured under prompts eliciting hundreds of output tokens, and their own Table 3 shows that fraction collapsing to 0.04 for short answers. Shin et al. [10] decompose sub-3B VLM latency by component on Jetson hardware and report that theoretical reductions do not predict on-device cost. We adopt their decomposition and reach the same conclusion by a different route in §3.2.

3 Setup

3.1 Configurations and oracle

For each example $(I, q, a^\star)$ we run a set $\mathcal{C}$ of visual configurations and record, for every $c \in \mathcal{C}$, the answer, its correctness, the model's per-step generation scores, and hardware measurements. Configurations comprise a blind baseline, low-resolution previews at two sizes (Answer-Low), a resolution-capped full pass, one preview+crop per cell of a 2×2 grid (Crop), and preview+OCR transcript variants (Ocr). The full pilot is 1000 examples across VQAv2, TextVQA, DocVQA and V*Bench, giving 13 000 instrumented generations.

Correctness uses the metric each benchmark defines (VQA accuracy for VQAv2 and TextVQA, ANLS for DocVQA, exact match for V*Bench multiple choice) and is recomputed offline from stored answers, so changing the metric never requires re-running the model.

The oracle labels each example with the cheapest configuration that answered correctly, under

$$J(c) = w_e \mathbf{1}[\hat{a}_c \neq a^\star] + \lambda_t t_c + \lambda_e E_c + \lambda_m M_c + \lambda_v V_c, \quad (1)$$

with t latency, E energy above a measured idle baseline, M peak memory and V visual tokens. Weights in Eq. 1 are calibrated so each resource term contributes 0.03–0.09 against an error weight of 1.0: the oracle requires correctness first, then discriminates on measured cost.

What the weights can and cannot change. Calibrating them so each resource term lands in a narrow band is a convention, not an argument, and Proposition 1 minimises this function, so a reader is entitled to ask what a different choice would do. Two answers, one structural and one measured.

Structurally, **oracle accuracy cannot depend on the weights at all**. The oracle ranks only among configurations that answered correctly, so no setting of λ can make a wrong answer preferable to a right one, and the set of solvable examples is fixed before any weight is read. Sweeping every weight over four orders of magnitude confirms it exactly: accuracy stays at 0.694 to within 10^{-9}.

For the same reason w_e **is inert in the labelling**: it is added to options the oracle has already discarded, so 100% of labels survive a 10 000× change in it. The error weight matters only through the ratio $V = w_e/\lambda_t$ in the serving rule of §7, where it is swept rather than fixed. Our own calibration story, phrased as resource terms measured "against an error weight of 1.0", therefore describes something the labelling never consults.

What the weights do choose is *which* correct action is cheapest. A hundredfold rise in λ_t moves the oracle's mix from 16% to 34% Crop and from 11% to 2% Ocr, leaving 75% of labels unchanged. That is not a fragility, it is the budget-dependence this paper reports as a finding: when visual tokens are dear, a preview plus a transcript beats a high-resolution crop. The reader should treat every accuracy claim here as weight-free and every action-mix claim as conditional on a stated budget.

Two further properties of Eq. 1 are worth stating. Being linear in the resources, it expresses only finitely many policies: sweeping λ_v over its whole range on our pilot yields exactly two, Crop below a crossover and Answer-Low above. And a policy conditioned on a probe must be charged for the probe, so an escalated query pays both passes throughout this paper.

3.2 What a pass actually costs

Every policy comparison in this paper is a cost comparison, so the cost model has to be measured rather than assumed. We time each stage of one pass with explicit CUDA synchronisation, following the component decomposition of [10]. Two standard proxies turn out to mispredict, and both matter for what follows.

The vision encoder is half the visual cost. Table 2 attributes latency on three models spanning 8.6× in parameters and two distinct vision encoders.

model	encoder	extra prefill	encoder share
SmolVLM-256M	52.5	55.0	49%
SmolVLM-500M	52.4	53.5	50%
SmolVLM2-2.2B	193.6	252.8	43%

Table 2: Visual cost split at 768 px, in milliseconds.

This bounds a whole family of methods. Post-encoder pruning [4] can only recover the prefill half, because the encoder has already run when tokens are scored; reducing input resolution recovers both. **Token pruning is capped at roughly half the available saving**, and the ratio holds across the parameter range. It also tells us what a probe reading mid-prefill can and cannot save, which §5 makes precise.

Decode latency follows depth, not size. SmolVLM2-2.2B emits the same three-token answer in 47.2 ms against 77.0 ms for the 4.4× smaller 500M model, because it has 24 layers rather than 32; per-layer cost is roughly constant at 2.0–2.4 ms. Choosing an edge model by parameter count optimises the wrong variable, and a cost model linear in parameters, as the power fit of [8] is, will not transfer here.

Cache reuse cannot refund the cheap pass. Our accounting charges an escalated query for both passes with no reuse, and the obvious objection is that a serving stack would not. VLCache [17] reports 1.2–16× time-to-first-token speedups by reusing encoder and KV caches across multimodal requests. Two facts close this, and neither needs an assumption about what a stack achieves.

First, escalation is a cache miss by construction: VLCache keys its cache on a hash over the input pixels, and escalating submits different pixels. That literature refunds *repetition*, and escalation is the opposite of repetition.

Second, and independently, decode is uncacheable. Reading output entropy requires the cheap pass to *generate*; generation produces the answer, differs per query, and no cache returns it. Writing c for the fraction of encoder and prefill a cache refunds, the probe's saving per escalated query is $(1-c)(1-\ell/L)\,t_{\text{prefill}} + t_{\text{decode}}$, which runs from 103.1 ms at $c=0$ to a floor of 64.7 ms at a perfect cache, still 63% of the uncached saving. No refund fraction erases it.

Energy is not reported. Our NVML integration disagrees by 18–96% between configurations spending identical visual tokens, while their latencies agree to 4–5%; a standing check refuses to report joules until that is repaired. This also bears on reading [8], whose headline, decode dominates 86–97% of inference energy, is measured under prompts eliciting hundreds of output tokens. Their own Table 3 shows the decode fraction collapsing to 0.04 under short-answer prompts, the regime every dataset here occupies. All costs in this paper are therefore latencies.

3.3 The three routing questions

Let y_{cheap} and y_{full} be the correctness of the low-resolution and full-resolution passes. Three distinct binary targets can be formed over the same examples:

$$Q_1 : y_{\text{cheap}} \qquad \text{(is the cheap pass correct?)} \qquad (2)$$

$$Q_2 : \neg y_{\text{cheap}} \wedge y_{\text{full}} \qquad \text{(does escalation recover?)} \qquad (3)$$

$$Q_3 : \textstyle\bigvee_{c \in \mathcal{C}_{\text{route}}} y_c \qquad \text{(answerable at all?)} \qquad (4)$$

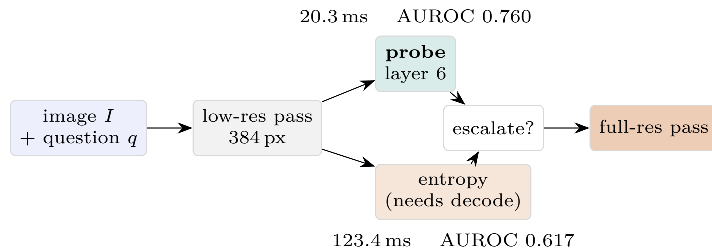

Figure 2: Two allocation signals read from the same low-resolution pass. Output entropy requires the pass to complete and generate; the probe reads the residual stream at layer 6 of 32 and needs no decode.

target	output entropy	pre-generation probe
Q_1 correct	0.758	0.822
Q_2 escalation helps	0.617	**0.760**
Q_3 answerable	0.539	0.681

Table 3: AUROC by target. Entropy is scored with the sign that favours it; read with the Q_1 sign it scores 0.372 on Q_2, because the sign flips between the two questions.

The escalation literature conditions on Q_1. The quantity that decides whether compute is well spent is Q_2, evaluated on the subset where the cheap pass failed. Q_3 separates queries no visual operation can rescue.

4 Confidence answers the wrong question

Mean token entropy over the generated answer predicts Q_1 well and Q_2 poorly. Table 3 reports AUROC on held-out folds, $n = 1000$ overall and $n = 617$ for Q_2.

The sign flip is the mechanism. A *confident* wrong answer is a knowledge failure that more pixels do not fix; an *uncertain* wrong answer is often a perception failure that they do. Splitting the 617 failures at their median entropy, escalation recovers 25% of confident failures and 42% of uncertain ones.

The two effects reinforce for a policy. Writing the net benefit of escalating as

$$G = \mathbf{1}[\neg y_{\text{cheap}} \wedge y_{\text{full}}] - \mathbf{1}[y_{\text{cheap}} \wedge \neg y_{\text{full}}], \qquad (5)$$

$\mathbb{E}[G]$ rises monotonically across entropy quintiles, from +5% to +41.5% (Fig. 3). Escalation is worth eight times more in the top quintile than the bottom.

Escalation is also not monotone: 4.1% of queries are answered correctly at 384 px and wrongly at full resolution. No surveyed method models this risk.

5 A pre-generation probe

Following [5, 6], we read the residual stream $h^{(\ell)} \in \mathbb{R}^d$ at the last prompt position (after the instruction, before

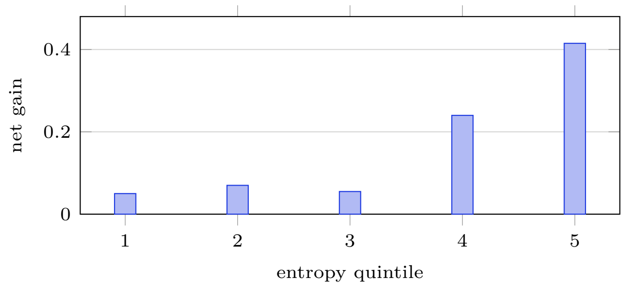

Figure 3: Net benefit of escalating, by entropy quintile ($n = 1000$, 200 per bin). AUROC of entropy for predicting net benefit is 0.734.

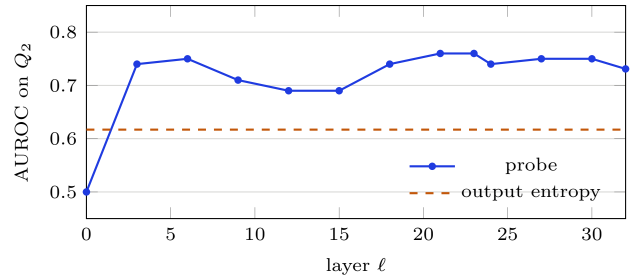

Figure 4: Probe AUROC on Q_2 against depth. The ceiling is reached by layer 3 and never improves, so the escalation decision can be made after one sixth of the language model.

probe family	pooled mixture	within DocVQA
difference of means	**0.749**	0.576
logistic	0.679	0.570
random Fourier features	0.699	0.596
shrunk centroid (LDA)	0.701	0.579
output entropy	0.735	**0.656**

Table 4: AUROC on the joint escalation target, 20 resampled folds, layer 6. Random Fourier features give a smooth non-linear boundary without a bespoke architecture; the shrunk centroid is the covariance correction the plain difference of means omits.

any token is generated) and fit the difference-of-means direction

$$w^{(\ell)} = \mu_+^{(\ell)} - \mu_-^{(\ell)}, \qquad s(h) = \frac{(h - \bar{\mu}^{(\ell)})^\top w^{(\ell)}}{\|w^{(\ell)}\|}, \tag{6}$$

where $\mu_\pm$ are class centroids and $\bar{\mu}$ their midpoint. Eq. 6 is deliberately linear: the question is whether the signal is linearly accessible, not how well a classifier can be tuned. Stated that way it is a framing rather than a defence, so we measured what capacity buys (Table 4).

Two things follow, and both are load-bearing. On the mixture the two-centroid difference *beats* every fitted alternative by 0.048 or more, so the paper's probe is not an undertrained stand-in for something better; fitting parameters on 700 examples in 960 dimensions costs more than it recovers. And within a domain no family closes the gap to output entropy: the best of the four reaches 0.596 against entropy's 0.656 $[0.645, 0.666]$. **The within-domain negative is a property of the representation, not of linear models**, which is the objection this table exists to answer. It remains one family of non-linearities rather than a proof, but it is the family a reviewer would reach for first. Fitting is a single pass over cached activations, and the resulting score is the one consumed by the decision node of Fig. 2.

Depth. Fig. 4 sweeps ℓ. The signal is absent at $\ell = 0$ by construction, the last prompt token is textual and has attended to nothing, and reaches its ceiling by $\ell = 3$. Bootstrapped, layer 6 gives 0.760 $[0.668, 0.849]$ and layer 23 gives 0.760 $[0.652, 0.853]$: indistinguishable.

Cost. Decomposing one pass with CUDA synchronisation around each component, a 384 px pass costs 11.0 ms in the vision encoder, 48.1 ms in prefill and 67.8 ms in decode. Reading at layer ℓ needs the encoder plus a fraction of prefill,

$$t_{\text{probe}}(\ell) = t_{\text{enc}} + t_{\text{proj}} + \tfrac{\ell}{L}\, t_{\text{prefill}}, \tag{7}$$

giving 20.3 ms at $\ell = 6$ against 123.4 ms for a complete pass.

That ratio is *not* the saving, and it is worth being explicit about why. Entropy requires the cheap pass to finish, but so does answering, whenever the policy decides *not* to escalate. The probe therefore saves only on escalated queries, where the remainder of the cheap pass is abandoned. With escalation rate ρ and full-resolution cost t_{full},

$$C_{\text{entropy}} = t_{\text{cheap}} + \rho\, t_{\text{full}}, \tag{8}$$
$$C_{\text{probe}} = (1-\rho)\, t_{\text{cheap}} + \rho\, \big(t_{\text{probe}} + t_{\text{full}}\big). \tag{9}$$

At our observed $\rho \approx 0.30$, Eq. 9 gives 185.3 ms against 154.3 ms: a 17% saving, rising to 31% if every query escalates and falling to 7% at $\rho = 0.10$. The probe is cheaper, but by a sixth, not by the 6× that comparing t_{probe} from Eq. 7 against a full pass would suggest. We state this explicitly because the read-cost ratio is the number an efficiency paper is tempted to report.

5.1 A better ranking is not a better policy

AUROC says a signal orders queries well; it does not say a policy built on it serves better answers per unit of compute. Building both policies at matched escalation rates and charging them by Eq. 9 exposes a trap.

A probe trained on Q_2 (the *conditional* target, defined only on queries the cheap pass failed) produces a policy that is 8.5 accuracy points *worse* than the entropy threshold it beat on AUROC (-0.085 $[-0.130, -0.040]$, paired). The reason is that a deployed policy must decide on every

policy (escalating 30%)	accuracy	latency	AUROC on G
always cheap	0.385	123.4	n/a
entropy threshold	0.480	182.2	0.751
probe on Q_2	0.395	151.7	0.259
probe on G	**0.510**	**154.3**	**0.761**
always full	0.530	206.0	n/a
oracle	0.580	139.6	n/a

Table 5: End-to-end policy comparison on the held-out fold, latency in milliseconds. Against the entropy threshold, the joint-target probe differs in accuracy by $+0.030$ $[-0.005, 0.070]$, indistinguishable, and in latency by -27.8 $[-40.7, -15.0]$ ms, a significant 15% saving.

policy	escalates	accuracy	latency
always cheap	0%	0.385	123.4
self-report, "ANSWER or ZOOM"	7%	0.385	137.9
self-report, "detailed enough?"	22%	0.405	169.8
entropy threshold	20%	0.465	164.7
probe	20%	**0.475**	**144.0**
always full	100%	0.530	206.0

Table 6: Untrained self-report against uncertainty-based routing, 200 held-out examples, latency in milliseconds. Against the stronger self-report the probe is $+0.070$ $[0.020, 0.120]$ more accurate and 25.8 $[11.9, 39.2]$ ms cheaper, both paired and both significant.

query, not only on failures, and the quantity it needs is the joint event G: escalation helps only when the cheap pass would fail *and* full resolution would succeed. Entropy predicts G at 0.751 because both factors rise with it. The Q_2-trained probe predicts G at 0.259, far below chance, because it never saw a successful cheap pass.

Retraining the same probe on G fixes it (Table 5).

And the Q_2 row is the more useful result: a probe can beat a baseline on a conditional target and lose decisively as a policy. Papers that report ranking quality without building the policy risk exactly this.

Asking the model directly is worse than reading its uncertainty. The escalation mechanism published systems train is a self-report: shown a downsampled image, the model states whether it needs the full-resolution one. We cannot reproduce VisionThink's GRPO at sub-1B, that is the premise of this work, but we can implement its *mechanism* without the training, and ask what the untrained version is worth. Two phrasings were tried, both eliciting a one-word judgement (Table 6).

The first phrasing escalates 7% of queries and recovers nothing: its accuracy is exactly that of never escalating, so the compute is pure waste. The second escalates 22% and buys two accuracy points, where routing on entropy at the same rate buys eight and routing on the probe buys nine at lower cost. The self-report is not merely weaker; it escalates as often as the alternatives and selects the wrong

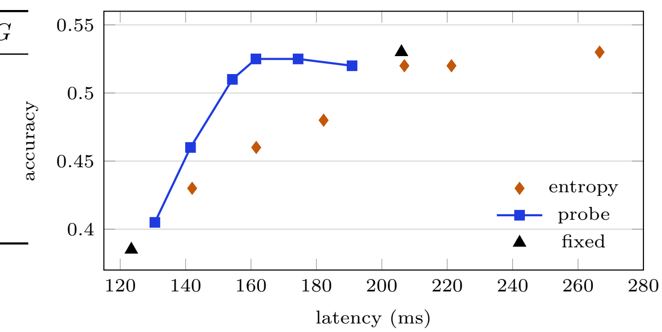

Figure 5: Accuracy against latency as the escalation rate sweeps from 10% to 70%, on the reported fold. Four probe points lie on the Pareto front, together with the two fixed policies at its ends. Fig. 6 tests how much of this survives resplitting.

queries.

This bounds what the untrained mechanism delivers and, read the other way, quantifies what the training in [1, 2, 3] is buying. It is not a comparison against those systems, which would require reproducing their training at a scale where it is not affordable.

The whole frontier, not one operating point. A single escalation rate flatters or damns arbitrarily. Sweeping it (Fig. 5) gives a cleaner statement: on this fold every entropy operating point is dominated. Of the fourteen policies evaluated, six lie on the accuracy–cost Pareto front, four of them probe-based and none entropy-based.

At matched accuracy the saving is 12% at 0.46 and 22% at 0.52. The 0.52 point matters most: the probe reaches 0.525 at 161.5 ms where always running at full resolution reaches 0.530 at 206.0 ms, within half a point of the accuracy ceiling for 78% of its cost, while the entropy policy at the same accuracy costs 206.9 ms, marginally *more* than simply always escalating.

How much of that is the fold? Universal dominance is a statement about six points on 200 test examples, and this project has been burned once already by a conclusion drawn at that scale. We therefore refit everything downstream of the split (probe direction, thresholds, calibration) on 200 stratified resplits.

The universal claim does not survive: all seven entropy operating points are dominated in only 52% of splits. What survives is stronger than that number suggests. On average 0.83 of seven entropy points escape domination, and when one does it wins by a median of 0.010 accuracy, two queries out of two hundred, with 85% of survivors winning by 0.02 or less. Meanwhile the probe holds 90% of the Pareto front, and at matched accuracy it is cheaper in 200 of 200 splits, by a median of 27.8% (Fig. 6).

The honest summary is that the direction of the effect is robust and its universality is not. We report the distribution rather than the fold because the fold is what a

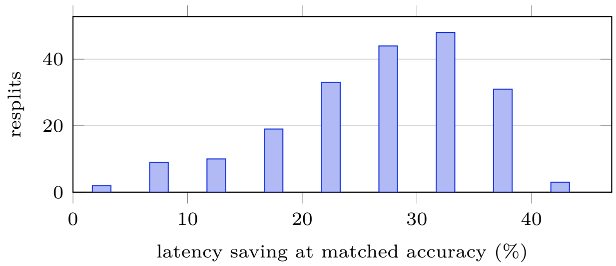

Figure 6: The probe's latency saving over entropy at matched accuracy, across 200 stratified resplits. The distribution is entirely positive; the single-fold figure of 12–22% quoted above is a conservative corner of it.

single split of 1000 examples can support.

A better ranking, again, and again no better policy. Since our own numbers say the ceiling is the signal, we tried to improve the signal. Entropy and the probe are complementary: a logistic combination of the two reaches 0.805 on the joint target, against 0.751 and 0.761 alone, a gain of $+0.054$ $[-0.010, +0.120]$.

As a policy it is strictly worse than the probe alone (0.505 accuracy at 181.1 ms, against 0.510 at 154.3 ms) because reading entropy requires finishing the cheap pass, which is precisely the cost the probe existed to avoid. The best ranking and the best policy are different objects here, and the second time we observed this it was not a mistake but a structural feature: a signal's value to a policy is bounded by what it costs to obtain.

How far from the oracle? Not close, and the shape of the gap is itself informative. The oracle escalates only 20% of traffic and is simultaneously more accurate *and* cheaper than always running at full resolution (0.580 at 139.6 ms against 0.530 at 206.0 ms). Measured against the always-cheap baseline, the joint-target probe closes 64% of the accuracy gap but *increases* cost, where the oracle decreases it. No policy we build moves along the direction the oracle occupies.

This is the strongest evidence in the paper that the ceiling is the signal rather than the policy, and it agrees with what the routing literature reports for text: NVIDIA's prefill-activation router closes 45.6% of its oracle gap [7], and [5] conclude that routing effectiveness is limited by the reliability of the success estimates, not by the routing rule.

The direction is not domain-general. Training on detail-limited datasets (DocVQA, V*Bench) and testing on knowledge-limited ones (VQAv2) gives 0.422; the reverse gives 0.366. Both fall *below* chance: the direction inverts rather than degrading. The 0.760 result holds only because the training fold spans all four datasets. "Would more pixels help" is not one internal quantity but

signal	pooled	within one dataset
image size (free)	0.748	0.534
output entropy	0.751	**0.664**
pre-generation probe	**0.761**	0.519

Table 7: Predicting the joint escalation gain. Pooled is the held-out fold of the four-dataset mixture, $n = 200$. Within-dataset trains and tests the probe inside a single dataset over 40 resamples, weighted by dataset size.

at least two (one for answers present but unreadable, one for answers absent) represented along opposing axes.

6 What the probe is actually reading

A cost measurement, not a literature review, exposed this. Pricing escalation per example rather than per configuration showed the probe systematically escalating *larger* images than entropy does, 1.19 to 1.33× the visual tokens at matched rate. That is worth explaining, and the explanation undermines the result above.

The flat escalation price was wrong. Our profiling image had a longest side below 1536 px, so the processor capped it and the "full-resolution" pass spent 320 visual tokens. On the pilot, Crop-free full passes average 424 tokens and exceed the 768 configuration on 44% of examples. Latency is affine in token count ($100.9 + 0.326\,v$ ms, worst residual 1.7 ms over four profiled configurations) so the honest per-example price is 238.9 ms against the 206.0 we charged, 16% higher. This is the third measurement error of the same family we have found, after processor upscaling and patch-grid quantisation, and all three come from a configuration name not determining a cost.

Under honest costing the probe's saving over entropy shrinks from -27.8 $[-40.7, -15.0]$ ms to -11.8 $[-27.2, +5.0]$ ms at a 30% escalation rate: no longer distinguishable from zero. It survives at 20% and 40%.

Why a better signal picks dearer escalations. The queries more pixels help are the queries with more pixels to add. A signal that ranks escalation value therefore also selects the expensive tail, and ranking quality partly cancels against cost saving. This is a fourth way a better signal fails a policy, and unlike the first three it is invisible unless cost is measured per query rather than per configuration.

And the correlation runs deeper than cost. If image size predicts escalation value, it is a routing signal in its own right, and a free one, read from the file header before any model runs. Table 7 scores it against the two signals the paper has been comparing.

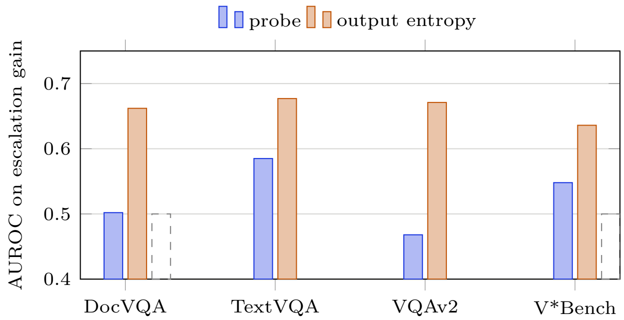

Figure 7: Within a single dataset, trained and tested inside it. The dashed line is chance. The probe's pooled advantage does not survive removal of the between-domain axis; entropy's largely does.

	DocVQA	TextVQA	VQAv2	V*Bench
test fold	30%	25%	30%	15%
probe favours	**50**%	36%	12%	2%
entropy favours	**0**%	28%	46%	26%

Table 8: Dataset composition of the 50 queries where each signal most wants to escalate relative to the other. Not one of the queries entropy prefers is a DocVQA page.

The probe's margin over a feature costing nothing is +0.013. And the within-dataset column is the result (Fig. 7): with the between-domain axis removed the probe is at chance in every dataset (DocVQA 0.502 $[0.483, 0.521]$, VQAv2 0.468 $[0.422, 0.513]$) while output entropy retains 0.664. Each of these fits sees only 210 training examples, which is a confound of its own; §6.1 removes it with a pilot built for the purpose.

What the disagreement looks like. The statistics above say the probe tracks image size; the queries say what that means. Ranking the held-out fold by each signal and taking the largest rank disagreements, the two sets are almost disjoint by dataset (Table 8).

The disagreement between the two signals predicts which dataset a query came from at AUROC 0.738, which is most of what either signal achieves on the escalation target itself. Where they differ, they differ about what kind of image it is.

The individual queries are unambiguous. The four the probe most wants to escalate are all DocVQA, all 448–640 visual tokens at full resolution, and all have $G = 0$: "What is written in big letters on the top right?" is answered *correctly* at 384 px and the escalation changes nothing. The four entropy most wants to escalate are all VQAv2 photographs at 320 tokens, all wrong at both resolutions, questions like "What is tucked behind the lamp?" where the model is genuinely lost and more pixels do not help.

Read charitably this is the probe doing something reasonable, since documents do repair more often on average. Read as a routing signal it is a category detector: it escalates a correct answer on a large page and declines a hopeless one on a small photograph, and neither decision used anything about the query.

The text-probe literature saw this coming, and evaluates accordingly. Having found it, we went back to check whether the work we build on is exposed to the same thing. It is not, and one of them names the mechanism. Moreno Cencerrado et al. [6] train and test their correctness direction *within* each dataset and report cross-dataset transfer separately, and observe that for some sources "the learned direction captures dataset-specific cues correlated with correctness rather than correctness itself". Lugoloobi et al. [5] likewise average AUROC over their benchmarks rather than pooling examples across them. Neither pools; we did.

Two of their observations also validate the test we use here. Cencerrado et al. report that probe performance on GSM8K "plateaus near random chance regardless of dataset size", distinguishing an absent signal from an under-sampled one by exactly the learning-curve argument, and they find a non-linear classifier on external embeddings generalising *worse* than a linear probe on internal states, the opposite of what a capacity explanation would predict. The correction here is therefore to our evaluation protocol, not to their finding.

This explains a result we had already reported and misread. We found earlier that the probe's direction inverts across domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) and read it as a transfer limitation calling for a domain-representative calibration set. The correct reading is stronger: a direction that carries no within-domain signal and inverts between domains is a domain classifier, and its pooled AUROC measures how separable the mixture is, not how well the model knows its own needs. Both facts were in our data; only the second explains them.

6.1 A single-domain pilot, and what it settles

The within-domain fit above gets 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. We said so, and then collected the pilot that decides it: 1200 DocVQA pages, enough to fit and test inside one domain.

The decisive form is a learning curve. If the probe were merely starved, AUROC would climb toward the pooled figure as training data grows. Neither extreme is what happens (Fig. 8).

The probe does improve, by +0.050 as training data grows ninefold, so the collapse is not purely starvation. But it plateaus far below output entropy, which is fitted on nothing. **The signal is genuinely weaker inside a domain than the pooled figure suggests**, and the

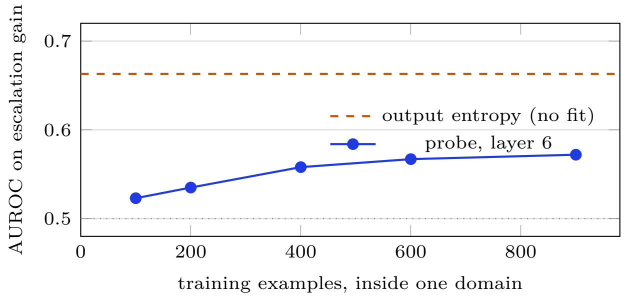

Figure 8: Probe AUROC inside DocVQA against training size, 30 resamples with a fixed 300-example held-out set. The dotted line is chance. More data helps and does not close the gap: at 900 examples, more than the pooled fit that produced 0.761, the probe reaches 0.572.

confound conclusion stands in this corrected form rather than as "at chance".

And the depth that works does not transfer either. We selected layer 6 because AUROC saturates there on the mixture, and the entire read-cost argument of §5 rests on stopping a sixth of the way through the language model. Inside one domain the signal keeps improving with depth:

layer	1	3	6	12	20	32
AUROC	0.531	0.546	0.572	0.579	0.618	**0.625**

The best depth is the last one. That costs the probe its cheap read: Eq. 7 gives 20.0 ms at $\ell = 6$ and 59.1 ms at $\ell = 32$, so the saving per escalated query falls from 106.9 to 67.8 ms.

Two things are worth noting about that residual 67.8 ms. It is exactly the decode time, because a full-prefill probe still skips generation, and it is the same floor the cache analysis of §3.2 arrived at from an unrelated direction. The probe's advantage narrows to one mechanism, stated twice: reading the model without making it speak.

An independent corroboration of the corrected depth. GlimpsePrune [4] reads its learned glimpse token at layer $K = \frac{2}{3}L$, chosen as the point where it has "aggregated sufficiently clear importance information" at a favourable overhead trade-off. That is where our within-domain sweep becomes informative (0.618 at layer 20 of 32) and far from the layer 6 the mixture selected. Two groups reading different internal signals for different purposes land on the same region of the network, which is weak evidence but it points the same way as §6.1: the shallow saturation was the artefact, not the deep one.

What survives. Three things. The mixture result is real and mixtures are real deployments: a router serving documents and photographs through one endpoint faces exactly the pooled distribution, and there a free image-size feature is nearly as good and much cheaper than either signal we studied. The negative results are untouched, since none of them rests on the probe working. And the paper's thesis survives inverted: we set out to show that a better signal is not a better router, and the sharpest instance turns out to be our own signal, whose advantage was an artefact of the evaluation mixture.

What does not. The claim that a 500M VLM encodes its need for more pixels in a linearly readable direction. On the evidence here it mostly encodes what kind of image it is looking at, which correlates with that need across a benchmark mixture and much less within one. §6.1 is the test we said this needed, run on a corpus built for it, and it does not rescue the claim: given nine times the training data the direction reaches 0.572 where the model's own output entropy, fitted on nothing, reaches 0.663. The honest statement is that the direction carries some within-domain signal and less than the cheaper alternative.

7 The escalation rate is not a free parameter

Every policy above picks a rate and thresholds at that quantile. That is a tuning exercise, and nothing connects the chosen rate to Eq. 1, the cost function the oracle is already labelled under. Kotte's UCCI [11] closes the same gap for text cascades (calibrate the signal to a probability, then let the cost model choose the threshold) and its transposition here is short, but it needs one correction.

The rule. Escalating x changes expected cost by

$$\Delta J(x) = -\, w_e\, \mathbb{E}[G \mid x] + \lambda_t\, \Delta t, \tag{10}$$

so the optimal action is to escalate exactly when

$$\mathbb{E}[G \mid x] \;>\; \tau = \frac{\Delta t}{V}, \qquad V = \frac{w_e}{\lambda_t}, \tag{11}$$

where V is the only knob: the latency an operator will spend to buy one additional correct answer, in milliseconds. Following [14], we state the trade-off in the units of the outcome rather than as a weight per millisecond, because the latter is not readable across devices. Estimating $\mathbb{E}[G \mid x]$ from the probe score by isotonic regression (monotone, non-parametric, one pass over cached activations) gives Algorithm 1.

Why isotonic, and what actually matters about it. We take isotonic regression from [11], and borrowing a component is how a paper inherits someone else's assumptions. The settings differ where calibration is sensitive: their target is a probability with tens of thousands of calibration points, ours a signed gain with a few hundred. Three families thresholded at the *same* break-even, so

Algorithm 1 Calibrated gain-optimal escalation

fit
1 read $h^{(\ell)}$ at the last prompt position, $\ell = 6$
2 $G_i \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^{\text{full}}] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^{\text{full}}]$
3 $w \leftarrow \mu_+ - \mu_-$ over $\{G_i > 0\}$ vs. the rest
4 $s_i \leftarrow (h_i - \bar{\mu})^\top w / \|w\|$
5 $\hat{g} \leftarrow 2\,\text{isotonic}\big(s,\ (G+1)/2\big) - 1$
serve x, given V
6 $\Delta t \leftarrow t_{\text{probe}} + t_{\text{full}} - t_{\text{cheap}}$
7 **if** $\hat{g}(s(x)) > \Delta t / V$ **then** abandon the prefill, run full
8 **else** finish the cheap pass and answer

calibrator	AUROC	calib. error	accuracy
isotonic	0.660	0.059	**0.736**
Platt sigmoid	**0.667**	0.098	0.713
equal-mass bins	0.656	**0.051**	0.733

Table 9: Calibrator families on 1200 DocVQA pages, 30 resamples, all thresholded at $\tau = 0.243$. Paired against isotonic, Platt loses 0.022 $[-0.026, -0.019]$ accuracy and equal-mass bins differ by -0.003 $[-0.006, 0.000]$.

any difference is the calibrator's magnitudes rather than a different operating point (Table 9).

The sigmoid ranks marginally *best* and calibrates worst, at nearly twice the error, and its policy is significantly less accurate. That is the paper's recurring pattern arriving in a third place: ranking quality does not transfer, this time because a rule that thresholds a magnitude is broken by a miscalibrated magnitude in a way a rank threshold would never notice. A parametric squash reports gains the operator did not ask for, and the rule lands somewhere other than the point V specifies.

Equal-mass bins, meanwhile, are indistinguishable from isotonic. **The load-bearing property is being non-parametric, not being isotonic**, and we say so rather than defend a component we happened to inherit. Isotonic remains the better default because monotonicity is free information that binning discards, but nothing here rests on it.

When the rule is optimal, and how it relates to UCCI's theorem. Eq. 11 is not a heuristic. Let $s(x)$ be any routing score and let Π_s be the policies measurable with respect to it, which is what a deployed router is: it sees the score, not the outcome.

Proposition 1. *Under the cost of Eq. 1 with $w_e, \lambda_t > 0$, and given the calibrated gain $\hat{g}(s) = \mathbb{E}[G \mid s]$, the policy*

$$\pi^\star(x) = \mathbf{1}\big[\hat{g}(s(x)) > \Delta t / V\big], \qquad V = w_e / \lambda_t,$$

minimises expected cost over Π_s. If $\hat{g}$ is non-decreasing, it is a threshold on s itself.

Proof. Write $C(\pi) = C_0 + \mathbb{E}\big[\pi(x)\,\Delta J(x)\big]$, where C_0 is the cost of answering cheap on every query and ΔJ is Eq. 10. The expectation is linear in π and the constraint set is $\pi(x) \in \{0, 1\}$, so it is minimised pointwise by escalating exactly where $\Delta J(x) < 0$. For $\pi \in \Pi_s$ the tower property gives $\mathbb{E}[\Delta J \mid s] = -w_e\,\mathbb{E}[G \mid s] + \lambda_t \Delta t$, so the pointwise test becomes $\hat{g}(s) > \lambda_t \Delta t / w_e$. Monotone $\hat{g}$ makes the sublevel set an interval in s. □

rule	calibration target	escalates	accuracy	latency
UCCI [11]	P(cheap wrong)	54%	0.520	178.5
ours	$\mathbb{E}[G \mid x]$	**34%**	0.520	**157.9**

Table 10: Both rules read the same probe score at $V = 800\,\text{ms}$ per correct answer and differ only in what they calibrate. Paired, the accuracy difference is 0.000 $[-0.025, 0.025]$ and the latency difference -20.6 $[-26.8, -14.9]$ ms: the same answers for 59% relatively more compute.

Two things follow. Isotonic regression returns a non-decreasing $\hat{g}$ by construction, so Algorithm 1 realises $\pi^\star$ whenever the calibration is accurate, which Fig. 9 measures rather than assumes. And the relation to Kotte's Theorem 1 [11] is exact: under their assumption that the escalated pass attains a fixed accuracy γ independent of which queries are sent, $\mathbb{E}[G \mid x] = \gamma - (1 - p_{\text{err}}(x))$, which is increasing in the weak model's error probability, so their rule and ours coincide. **The two rules agree exactly when escalation cannot make things worse, and Table 10 is what their difference costs when it can.**

The break-even is signal-dependent, and that is the probe's second advantage. Line 6 differences Eq. 9. A policy reading entropy must finish the cheap pass before deciding, so escalation adds the whole full-resolution pass, $\Delta t = 206.0\,\text{ms}$. A policy reading the probe abandons the cheap pass it never completed, so $\Delta t = 20.3 + 206.0 - 123.4 = 102.9\,\text{ms}$, exactly half. At the same operator preference V, the probe's break-even gain is half entropy's: **a cheaper signal is entitled to escalate on weaker evidence**, and the saving compounds rather than merely lowering the bill for a fixed rate.

Calibrating the wrong probability. UCCI's Theorem 1 assumes the strong model attains a fixed accuracy on the escalated subset, invariant to which queries are sent. Under that assumption the escalation value of a query is determined by the weak model's error probability, so calibrating P(cheap wrong) suffices. Visual escalation violates it: it is non-monotone, repairing 21% of queries and *damaging* 4%, so the escalated pass is a second random outcome rather than a fixed accuracy. The quantity to calibrate is the signed gain, and the difference is not academic (Table 10).

The failure is legible. A confidently wrong cheap pass has high error probability and near-zero expected gain, so UCCI escalates it and the gain rule does not, the same conditional-versus-joint error of §5.1, displaced from the

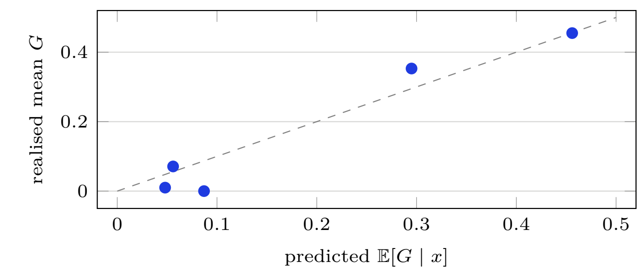

Figure 9: Reliability of the gain calibration on the held-out fold, binned by predicted gain. The dashed line is perfect calibration. A rule that thresholds a *magnitude* is only as good as the magnitude, which a ranking metric would not detect.

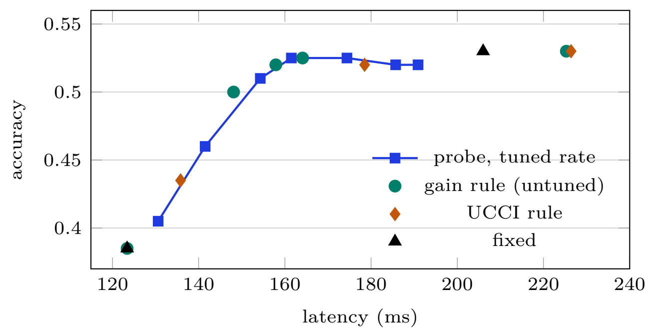

Figure 10: Untuned decision rules against the tuned sweep. The gain rule tracks the swept frontier without a rate parameter; the UCCI rule, calibrating correctness instead, sits inside it and collapses to always-escalate once V is large.

training target to the calibration target. Fitting the gain directly is well calibrated: binned over the held-out fold, predicted and realised gain agree to 0.029 (Fig. 9), against a correctness ECE of 0.027 for the same signal, so the correction costs nothing in calibration quality.

Does removing the knob cost anything? It should not, if the rule is right. Sweeping V over 100–3200 ms per correct answer places six operating points on the accuracy–latency plane without any rate ever being chosen; five of the six are undominated by the tuned sweep of Fig. 5 evaluated on the same fold (Fig. 10), and 78% remain undominated across the 200 resplits. An untuned rule reaches a tuned frontier, which is the useful form of this result: the rate was never information, only a stand-in for a preference the operator already has.

Where the rule refuses. At $V \leq 200$ the break-even of Eq. 11 exceeds any attainable gain and the rule escalates nothing, returning the always-cheap policy. That is the correct answer, not a degenerate one: by Eq. 10 no expected gain on this pilot repays 102.9 ms when a correct answer is worth 200. A tuned rate cannot express this, which is the failure mode reported for our earlier

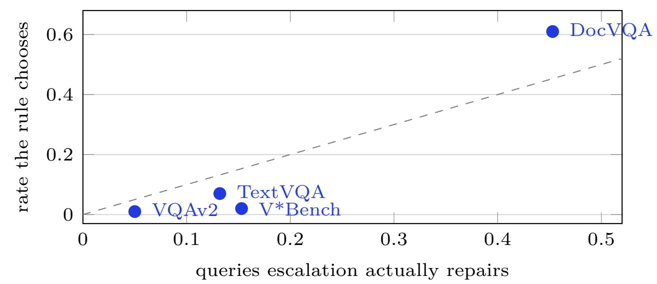

Figure 11: One value of V, applied unchanged to four domains. The rule escalates 61% of DocVQA and 1% of VQAv2, tracking how much escalation actually repairs there ($r = 0.974$). The dashed line is not a target, escalating every repairable query is not optimal, but it fixes the scale.

threshold tuners, they discovered "never escalate" on one fold and had no way to say whether that was a preference or an artefact. Where a floor on escalation is required instead, [12]'s recall-controlled gates give one that cannot degenerate by construction.

7.1 The rule transfers where the signal does not

Every number above uses the probe, which §6 discredits within a domain. So we re-ran the whole comparison on output entropy, the signal that does work within a domain, fitted and tested inside each dataset over 40 resamples, with escalation priced per example. If the rule is a property of the method rather than of the signal, it should survive.

It does, and the confound makes it more useful rather than less. Escalation value is domain-determined, so a fixed rate is wrong in every domain but the one it was tuned on. A rule calibrated on the observed gain distribution picks its own rate wherever it is fitted, from one unchanged operator preference (Fig. 11).

Fixing one policy and applying it to all four domains, weighted by dataset size, the rule reaches 0.489 accuracy at 181.6 ms where a tuned 30% rate reaches 0.464 at 195.1 ms: more accurate *and* cheaper. Tuned rates at 40% and 50% are dominated as well. Only the two lowest tuned rates survive on the front, at the cautious end where escalating little is right everywhere.

7.2 How far to escalate, not just whether

Every escalation method we have read is binary: answer from a downsampled image, or run the full one. Taking the cheapest configuration that answers each query correctly says that collapses the decision carrying most of the money.

Algorithm 2 Calibrated ladder escalation

fit, given rungs $r = 1 \ldots R$ in increasing cost
$s_i \leftarrow$ any routing score read from the cheap pass
for each rung r:
  $G_i^{(r)} \leftarrow \mathbf{1}[\neg y_i^{\text{cheap}} \wedge y_i^r] - \mathbf{1}[y_i^{\text{cheap}} \wedge \neg y_i^r]$
  $\hat{g}_r \leftarrow 2\,\text{isotonic}\big(s,\ (G^{(r)}+1)/2\big) - 1$
serve x, given V
$\Delta t_r \leftarrow t_r(x) - t_{\text{cheap}}(x)$ for each rung
$v_r \leftarrow V\,\hat{g}_r(s(x)) - \Delta t_r$
if $\max_r v_r > 0$ **then** run rung $\arg\max_r v_r$
else answer from the cheap pass

cheapest rung that answers correctly	share
384 px (no escalation)	38.3%
intermediate rung	**17.7**%
full resolution	5.2%
nothing we run	38.8%

Of the queries escalation can repair, 77% are repaired by the intermediate rung. A binary policy over-serves three quarters of what it escalates. The rungs are also not equally efficient: 384 → 768 buys 13.3 points for 79 ms (+1.69 points per second) where 768 → full buys 3.3 for 38 ms (+0.86). The binary jump measures +1.42, which is what averaging a good rung with a poor one gives.

The rule generalises without modification. Rung r has its own expected gain over answering cheap and its own extra latency, so

$$r^\star(x) = \arg\max_r \big(V\,\mathbb{E}[G_r \mid x] - \Delta t_r\big), \tag{12}$$

answering cheap when no rung is positive. Eq. 12 reduces to Eq. 11 when there is one rung; the binary policy is the special case that has deleted the middle of its own ladder. One calibrator per rung is fitted on the same score, so the additional cost is a second isotonic fit over cached values.

Proposition 1 extends without change. Escalating to rung r shifts expected cost by $\Delta J_r(x) = -w_e\,\mathbb{E}[G_r \mid x] + \lambda_t \Delta t_r$, the choice is again pointwise because the objective is linear in the selection, and the best action at x is the rung minimising ΔJ_r, or the cheap pass when every ΔJ_r is non-negative. Multiplying through by $-1/\lambda_t$ gives Eq. 12. The binary rule is therefore not a simplification of the ladder but a *constrained* version of it, forced to choose between two of the rungs available to it.

It pays where the ladder physically exists, and nowhere else. Paired over four domains and four preferences, the ladder is -9.6 $[-17.2, -2.6]$ ms cheaper at -0.003 $[-0.008, +0.002]$ accuracy: significantly cheaper, indistinguishable in accuracy. The per-domain breakdown is the real result (Table 11).

On DocVQA the ladder saves 35 ms at no accuracy cost. On TextVQA and VQAv2 it does nothing, correctly, because for those datasets there is no top rung to find.

	top rung exists	accuracy	latency
DocVQA	98%	+0.003	**−35.1**
V*Bench	98%	−0.008	−3.5
TextVQA	0%	−0.006	+1.2
VQAv2	0%	−0.002	−0.8

Table 11: Ladder minus binary, averaged over four operator preferences, latency in milliseconds. "Top rung exists" is the share of images on which the full-resolution pass spends more visual tokens than the intermediate one.

step	net gain [95% CI]	cost	points/s
64 → 320	+0.383 [0.352, 0.414]	105	**+3.65**
320 → 640	+0.086 [0.060, 0.111]	129	+0.67
640 → 1088	−0.002 [−0.026, 0.022]	156	−0.01

Table 12: Rung-to-rung value on 1200 DocVQA pages, by visual-token count, latency in milliseconds. Accuracy peaks at 640 tokens (0.748) and 1088 tokens reaches 0.746: overlapping intervals for 70% more tokens.

A configuration name does not determine a configuration. That last column is a measurement fact worth stating on its own. The processor caps its target at the input's longest side, so on a small image the "full-resolution" pass *is* the intermediate pass: identical token count, identical cost, identical output. This holds for 100% of TextVQA and VQAv2 and 2% of the other two, 56% of the pilot. The escalation this paper studies is a real escalation for half its data.

This is the fourth error of the same family we have found, after processor upscaling, patch-grid quantisation, and the flat escalation price of §6. All four come from treating a configuration label as a cost or a capability. The general form of the lesson: in adaptive-resolution work, verify that the configurations differ *on the data*, not in the config file.

A ladder built on measured rungs, and where it stops. The mixture cannot price the second and third rungs, because for over half of it they are the same configuration. So we collected a single-domain pilot designed against that: 1200 DocVQA pages, with rungs chosen by measuring SmolVLM's patch-grid buckets (64, 320, 640 and 1088 visual tokens) rather than by picking round pixel sizes. Every rung is strictly dearer than the one below it on 95–100% of images, and latency is affine in tokens at $205.0 + 0.415\,v$ ms with a worst residual of 20.2 ms.

The result is a saturation point, and it arrives early (Table 12).

Three readings. The first rung is *five times* more efficient per millisecond than the second, and the second is the last one that pays: the top step repairs 8.1% of queries and damages 8.3%, a wash costing 156 ms. Of everything escalation can repair, 93% is repaired below the top rung and only 4.2% needs it. And this is measured on document

	384	768	1152	2048	top step
SmolVLM-500M	0.279	0.662	**0.748**	0.746	−0.002
SmolVLM-256M	0.220	0.588	**0.653**	0.666	+0.013
SmolVLM2-2.2B	0.413	0.669	**0.738**	0.754	+0.016
median visual tokens spent at each rung					
SmolVLM-500M/256M	64	320	640	1088	
SmolVLM2-2.2B	81	405	**810**	1377	

Table 13: ANLS accuracy by rung, three models on the same 1200 DocVQA pages, columns labelled by the pixel target. Every top-step interval is a tight null: $[-0.026, +0.022]$, $[-0.012, +0.037]$ and $[-0.011, +0.041]$, half-widths 0.024 to 0.026.

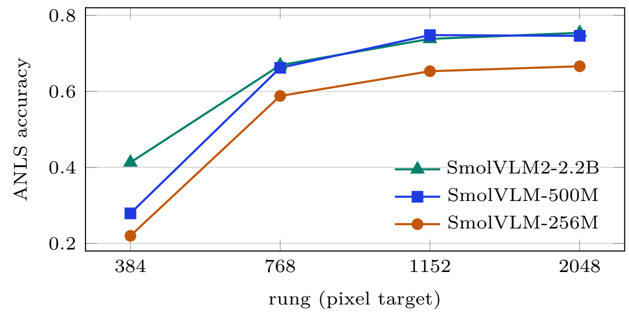

Figure 12: Accuracy against rung on the same 1200 DocVQA pages. The last step is flat for all three, though they spend 640, 640 and 810 visual tokens respectively at the rung where it flattens: the ceiling is a resolution, not a token budget.

pages, the regime where resolution matters most, so it is not a low-detail artefact.

More pixels stop buying accuracy well below the resolution the model accepts. That is a sharper form of the non-monotonicity in §3.3: escalation does not merely harm a few queries, it has a ceiling, and past it the harm rate catches the repair rate exactly.

A null result is only a result if it is tight, so it is worth saying that this one is. The interval on the top step has half-width 0.024, against measured gains of +0.383 and +0.086 on the steps below it: an effect a third the size of the smallest one we do detect would have been visible. That distinguishes this from a claim of no effect made at a sample size incapable of finding one, which is the failure mode we impose on ourselves elsewhere and check for here.

The ceiling belongs to the corpus, not to the model. A saturation point measured on one model could be that model running out of capacity, or the pages simply being legible at that resolution. The two explanations predict opposite things, so we re-ran the entire ladder on SmolVLM-256M over the same 1200 pages: capacity implies the smaller model saturates *earlier*, legibility implies the same rung with lower accuracy throughout (Table 13).

It is legibility. Halving the parameters costs accuracy everywhere and moves the ceiling nowhere, and all three top-step intervals are tight, so these are measured nulls rather than failures to measure. Plotted, the three curves rise and then flatten at the same place (Fig. 12).

And the ceiling is a resolution, not a sequence length. The two SmolVLM models share an 86M vision encoder, so "stops gaining at 1152 px" and "stops gaining at 640 visual tokens" name the same point for both and cannot be told apart. SmolVLM2-2.2B tokenises differently, spending 810 tokens where the others spend 640, which separates them. All three stop at the same *pixel* target while spending different numbers of tokens there. **What runs out is the information in the pixels, not the sequence length the model can use.**

V	accuracy delta	latency delta (ms)
400	+0.359 [+0.346, +0.374]	+86.0 [+84.3, +87.4]
800	−0.010 [−0.019, −0.001]	−161.1 [−163.9, −158.4]
1600	−0.010 [−0.017, −0.003]	−154.5 [−158.3, −150.7]
3200	−0.002 [−0.009, +0.005]	**−136.4 [−141.1, −132.0]**

Table 14: Ladder minus binary on 1200 DocVQA pages, paired within each operating point over 30 resamples. At $V = 400$ the binary rule barely escalates at all and the ladder buys 36 accuracy points for 86 ms; above that it trades at most one accuracy point for roughly 150 ms.

That makes the saturation point reusable. It is a property of the corpus, so it can be found once on the data and carried across a change of serving model and of vision encoder, which is a far more useful object than a threshold that must be re-derived per model. We note in fairness that the two smaller-gap models put a positive point estimate on the top step, leaning the opposite way from the 500M; at these interval widths that is noise, and we report it rather than claim a symmetry the data does not show.

The ladder rule finds the ceiling on its own. Given rungs and prices, Eq. 12 selects the top rung on 0–6% of queries across the whole preference range: it discovers the saturation from the calibrated gains without being told. Paired against the binary policy over 30 resamples of a held-out 300 (Table 14), it is significantly cheaper at three of four operating points, and at the highest preference it is cheaper at accuracy the comparison cannot separate.

The mixture measured this effect at −9.6 ms because most of its ladder did not exist. We report the intervals per operating point rather than pooled across them: an earlier version of this comparison bootstrapped the four preference means, which resamples four numbers and yields an interval four points wide.

A refinement of ours that does not pay. The cost audit suggested an obvious improvement: since escalation prices vary 2.5× across the pilot, charge each query its own break-even $\tau_i = \Delta t_i / V$ rather than a global one. Implemented and measured, it is worth nothing, +0.000 [−0.000, +0.000] accuracy and −1.5 [−4.3, +0.4] ms, both spanning zero.

The reason is the confound again. Image size is largely a *domain* property, so within a domain the prices a per-query rule would discriminate between barely differ, and there is nothing for the refinement to bite on. It would matter for a router serving one queue of genuinely mixed image sizes, and we report it as a negative rather than dropping it because the negative is the informative part: the same fact that makes the probe a domain detector makes per-query pricing redundant within a domain.

One caveat on the UCCI contrast. The gap of Table 10, calibrating correctness over-spending against calibrating the gain, is a mixture effect too. Within a domain both rules land on the front at different operating points, and the sign flip that separates them is weaker. We state the contrast as measured on the mixture and unresolved within a domain.

8 Knowing *whether* is not knowing *where*

Our measurements show region choice matters more than action choice, so we tested whether the same internal signals locate the region. SmolVLM lays its 64 visual tokens in a contiguous 8×8 grid, so the states covering candidate cell k can be pooled,

$$z_k = \frac{1}{|\mathcal{P}_k|} \sum_{p \in \mathcal{P}_k} h_p^{(\ell)}, \tag{13}$$

and ranked by a probe trained on which cells actually answered correctly. This requires no judge model, no reinforcement learning and no extra forward pass.

On a 2×2 grid the localizer scores 48–51% across six depths against 51% for choosing uniformly at random, with a 64% oracle ceiling. Overlapping cells make random choice strong there, so we repeated the experiment at 3×3 and then at 4×4. The finer grid is a fairer test: on the 400 examples all three share, the gap between random and the oracle widens from 12.2 to 16.2 points as cells shrink, so a localizer has more to find, not less.

One split is not enough to state a negative. Our earlier claim, that at no depth on either grid does the localizer beat random, rests on a single train/test split. Resampling it 60 times changes the sign of the answer (Table 15).

At 4×4 the localizer beats random at every depth, with intervals excluding zero. It also beats it by 0.9 accuracy points, which is 6% of the gap that a perfect localizer would close. **The honest claim is about magnitude, not sign**: the signal is real and it is too small to build on. We state it that way because we have insisted elsewhere in this paper that a null is only a result when the interval is tight enough to have found an effect, and the converse obligation applies here.

	2×2 ($n = 500$)	4×4 ($n = 400$)
depths beating random	1 of 6	**4** of **4**
best margin	+0.006	+0.009 [0.005, 0.014]
headroom to the oracle	0.122	0.162
share of the gap closed	n/a	**6**%

Table 15: Localizer margin over random cell choice, 60 resampled splits. Random is recomputed on each test fold rather than assumed, since the cell-correctness rate varies by fold.

family	configuration	tokens	accuracy
resolution	384 px	64	0.383
resolution	768 px	320	0.516
resolution	full	320	0.549
crop, random cell	2×2	**128**	0.392
crop, best cell	2×2	**128**	**0.584**

Table 16: Accuracy against visual tokens, $n = 1000$, two ways of spending the same budget. Paired against the best resolution configuration, the best crop is +0.035 [0.008, 0.063] more accurate at 2.5× fewer tokens.

What the localizer would have been worth. A negative result is easier to accept when the thing that failed is cheap. This one is not. Both ways of spending a visual-token budget are in the pilot, so they can be put on one axis (Table 16): a rung spends its tokens on the whole frame, a crop spends them on one cell at high magnification. This is also the comparison §3.2 has been making against the pruning literature [4] in words rather than numbers.

The two readings of the crop row are the whole point (Fig. effig:twoways). A *randomly* chosen cell reaches 0.392, which is what the same tokens buy in resolution (0.383 at 64, and about 0.42 interpolated at 128). The *best* cell reaches 0.584, beating the dearest resolution configuration while spending 2.5× fewer tokens. **Position is worth more per token than resolution, and only to a policy that knows where to look.**

That prices the failure. A working localizer would be the most token-efficient action in this action space, which is presumably why [3] is willing to spend cold-start fine-tuning, multi-turn GRPO and a 70B judge to obtain one. Our 0.9-point margin buys none of that headroom.

The low-resolution pass knows *whether* more resolution would help and essentially not *where* to spend it.

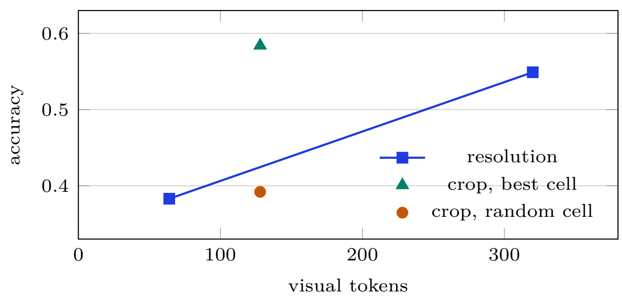

Figure 13: Two ways to spend a visual-token budget, $n = 1000$. The best crop sits above the resolution line at $2.5\times$ fewer tokens; a randomly chosen one sits on it. The vertical distance between the two crop markers is what a localizer would have to recover, and §8 recovers 6% of it.

This bounds what internal signals deliver at this scale, and suggests that the supervision machinery of [3] is not overengineering.

9 Limitations

Three serving models, one GPU, and two pilots: 1000 examples across four datasets and 1200 within one, the latter run on all three. The saturation ceiling is the claim we have pushed hardest, over an $8.6\times$ parameter range and two distinct vision encoders, and it has not moved. What we cannot say is whether it would move for a family trained differently: all three models come from one lineage and one data recipe, so a shared pretraining corpus remains a possible common cause. Our strongest baseline is the published escalation *mechanism* without its training; a re-implemented VisionThink at sub-1B scale remains the comparison the field would want, and the gap between an untrained self-report and a trained one is exactly what we cannot measure.

Four limitations deserve more than a mention. The latency saving needs an inference stack that can abandon a prefill mid-flight and start a different image. We verified this is real rather than arithmetic: physically truncating the decoder to its first six layers gives a 20.5 ms prefill, within 1% of Eq. 7's prediction, and an escalated query then costs 277.0 ms against 402.9 ms for completing the cheap pass first, a measured 31% saving. What standard serving frameworks do not expose is the *control* to do this per query, so the saving is available but not yet convenient. Absolute accuracies run from 0.385 to 0.580, a regime where the model is wrong about half the time, and conclusions drawn there may not transfer to a model that is usually right. The probe's direction does not transfer across question domains (trained on detail-limited data and tested on knowledge-limited data it scores 0.422, and the reverse 0.366, both below chance) so deployment needs a domain-representative calibration set and a shift detector. And the cross-model transfer result uses two models that share an identical 86M vision encoder, so part of what transfers may be encoder-side rather than genuinely cross-model; separating those requires a source model with a different encoder.

We ask many questions of one dataset mixture, so the intervals above are corrected rather than nominal. Thirteen paired comparisons at the nominal 5% level carry a 49% chance of at least one false positive, so we convert each bootstrap to a two-sided p-value and run Holm's step-down over the family. Nothing is lost: the nine comparisons that clear the nominal level are the same nine that survive correction, at adjusted $p = 0.0013$, because the surviving effects sit at the bootstrap's resolution floor while the failures are unambiguous nulls.

The split is itself informative. **Every surviving comparison is about cost**; all three probe-versus-entropy *accuracy* comparisons fail, as does the top rung's gain. That is the paper's claim structure certified rather than asserted: what we demonstrate is cheaper compute at accuracy the data cannot separate, not better answers.

The Pareto-dominance result is the one we have tested hardest and it came back qualified: universal dominance holds on the reported fold and in 52% of resplits, so the claim we make is the distributional one (90% of the front, a saving positive in 200 of 200 splits) and not the categorical one. Readers should treat any single-fold dominance statement in this literature, including our own earlier one, as untested until resampled.

One limit is specific to §7: V is elicited, not measured, so the contribution is that the policy is determined *given* a preference, not that we know the preference. The adversarial result is a proxy: it shows the probe is not incidentally moved by what moves entropy, not that it resists an attack optimised against it.

The confound audit of §6 has a limit of its own, in our favour and worth stating. Fitting inside one dataset gives the probe 210 training examples where the pooled fit gets 600, so part of the collapse could be sample size rather than absent signal. Two observations argue against that reading (entropy needs no training at all and is therefore unaffected, and the probe is at or below chance rather than merely noisy) but the clean test is a single-domain corpus large enough to fit and test inside, and 300 examples per dataset is not it. We report the confound as established and its magnitude as uncertain.

10 Reproducibility

Every numeric claim in this paper is re-derived from the run records by a validation harness with an explicit threshold per claim, which exits non-zero if any stops holding; it currently checks fifty-six, including the negative results on localization, on our own energy instrument, and on the universality of Pareto dominance, that last one asserts the claim is *not* universal, so restating it too strongly fails the build. Claims stated in the text and not covered by a check are marked as unverified. The oracle runs are stored as append-only JSONL with per-configuration answers, generation scores and hardware measurements, so correctness metrics and cost weights can be changed offline without re-running any model. Splits are hash-based on example identifiers, so growing the pilot never reshuffles examples already assigned.

Compute. Every experiment in this paper runs on one RTX 4060 laptop GPU with 8.6 GB of memory. The oracle runs comprise 54 660 instrumented generations totalling 305 minutes of measured model time, of which the 1200-page single-domain pilot is 6 000 generations per serving model, 40 minutes each for the two SmolVLM models and 54 for SmolVLM2-2.2B; activation extraction and the component profiles add roughly a further half hour. Wall-clock is appreciably larger, since image decoding, the processor, and the instrumentation itself sit outside the timed region. No training of any kind was performed: every probe here is a difference of means over cached activations, and every calibration a single isotonic pass, both of which take seconds on a CPU. The reproduction cost of this paper is therefore about six GPU-hours, which is the point of working at this scale.

11 Conclusion

We set out to ask what a 500M vision–language model knows about its own need for more pixels before it generates anything. The answer we can defend is narrower than the one we expected, and arriving at it was the useful part.

On a four-dataset mixture, a linear probe on pre-generation activations ranks escalation value better than output confidence and yields a policy readable after a sixth of the language model. Inside any one of those datasets it reaches 0.572 where output confidence, fitted on nothing, reaches 0.663, and more training data narrows that gap without closing it. What the direction mostly encodes is which *kind* of image is present: a quantity that predicts escalation value across a benchmark mixture, because the mixture's datasets differ enormously in how much resolution buys, and predicts much less within one. Raw image size, free to read, captures almost all of it.

Two things bound the claim rather than one. This is a way of deciding *whether* to look again and not *where*, and on the evidence here it is a way of deciding for a heterogeneous mixture and not for a workload.

What we would keep is the part that did not depend on the probe. The decision rule is signal-agnostic and survives on output confidence, where one stated preference produces a different escalation rate per domain without retuning. And the resolution ladder turns out to carry most of the money that was attributed to the escalate/answer decision: on document pages, 93% of what escalation repairs is repaired below the top rung, and the ceiling sits at a pixel resolution that three models with two vision encoders agree on.

The result we would most like others to carry away is the negative one in §5.1, which we encountered four times. A probe trained on the conditional target beat its baseline on ranking and lost the policy comparison by 8.5 points. An ensemble of entropy and probe then beat both on ranking and lost the policy comparison on cost. A rule calibrating the weak pass's error probability, which is the standard cascade recipe, matched our accuracy while spending 59% relatively more compute. And the signal that beat all of them on the mixture turned out to be ranking the mixture. The target was wrong, then the signal was too expensive to read, then the calibration target was wrong, then the evaluation distribution was doing the work. None of the four is visible in an AUROC table.

Reporting ranking quality without building the policy is reporting the wrong number. The four ways of getting it wrong are optimising a target the policy does not face, optimising a signal the policy cannot afford, calibrating a quantity whose assumptions the setting violates, and reading a between-group difference as a within-group one. All are easy to miss.

The last is the one we would flag most broadly, with the caveat that the literature we borrowed the probe from had already avoided it: any probe evaluated on a mixture of benchmarks that differ in the outcome being predicted should be scored within each of them before it is believed. The text-routing work reports per-dataset AUROC and warns explicitly about directions that capture dataset-specific cues [6]; we pooled, and pooling is what produced our headline. The check costs one line, and it was our own evaluation protocol rather than the method that failed it.

A What each part is worth

The components above are each justified where they are introduced, over five sections and on folds and cost models that were not always identical. That is how a paper acquires an inconsistency it cannot see, so we recompute all of them against one reference on one fold with one cost model.

Read at a fixed preference, an ablation says the opposite of the truth. Our first attempt held V constant and compared raw accuracy. Every removal appeared to *improve* accuracy, because every removal escalated more, and at fixed V nothing prevents a variant from buying accuracy with latency. That is the error this paper spends §5.1 warning about, committed on ourselves. Each variant is therefore swept to a frontier and read at a fixed latency budget (Table 17).

Three things. The losses are real but wildly unequal, so the table is a ranking of what matters rather than a list of things that help. **At 400 ms the ladder is what makes the budget usable at all**: binary escalation reaches

variant	acc. at 400 ms	acc. at 500 ms
reference	**0.653**	**0.743**
no per-example pricing	0.650 (−0.003)	0.744 (+0.000)
no calibration (tuned rate)	0.431 (−0.221)	0.571 (−0.172)
no ladder (binary)	0.281 (−0.372)	0.622 (−0.121)
no gain target (UCCI)	0.278 (−0.375)	0.611 (−0.133)
no signal (fixed policy)	0.278 (−0.375)	0.278 (−0.465)

Table 17: Each variant swept over six operator preferences and read at a fixed budget, 1200 DocVQA pages, 30 resamples. Deltas are paired against the reference frontier.

serving model	cheap acc.	full acc.	probe AUROC	entropy AUROC
SmolVLM-500M	0.383	0.549	0.761	0.751
SmolVLM-256M	0.298	0.511	0.682	0.616

Table 18: AUROC is on the joint escalation-gain target. On the reported fold of both models every entropy operating point is Pareto-dominated; six probe points lie on the front for the 256M model, four for the 500M. The resplit test of Fig. 6 was run for the 500M model only.

0.281, which is always-cheap to within 0.003, because no binary policy can afford the top rung inside that budget. And per-example pricing costs −0.003 [−0.004, −0.001]: an interval excluding zero, so a real effect, and small enough that we report it as negligible rather than as nothing. Those are different claims and the check that guards this table asserts the second.

B A second serving model, and cross-model transfer

Replication on a second serving model. Every result above has one model deciding for itself, so we repeated the entire chain (oracle run, activation extraction, probe fitting, rate sweep) with SmolVLM-256M as the model that answers, on the same 1000 images and questions (Table 18).

The finding replicates and strengthens. On the smaller model the probe's advantage on the joint target widens from +0.010 to +0.066, and at the highest escalation rate both signals reach 0.455 accuracy while the probe costs 190.3 ms against 278.0 ms, 32% less, against 12–22% on the 500M.

Two properties survive the halving of model size and one does not. Self-knowledge is unchanged: AUROC on "will the cheap pass be correct" is 0.751 against 0.758, a difference of seven thousandths for a model with half the parameters and 16 fewer accuracy points. The escalation *opportunity* also grows, escalation repairs 24% of queries against 21%, and damages 3% against 4%. But knowing whether escalation will help degrades, from 0.734 to 0.672: half of the smaller model's failures are unsolvable at any resolution, so the recoverability signal has less to work with. A weaker model needs routing more and can route slightly less well.

Cross-model transfer. Following the Encoder–Target Decoupling of [7], we predict the 500M model's escalation outcomes from SmolVLM-256M activations on the same images. This scores 0.785 $[0.692, 0.873]$ at layer 30 and 0.770 at layer 1: at least as well as the 500M model predicts its own. The probe can therefore run off the serving path entirely, on a separate model that can be batched or placed on different hardware.

perturbation	signal	shift	escalation
high-frequency noise	entropy	$+0.28\sigma$	50% → **62%**
high-frequency noise	probe	-0.04σ	50% → 50%
flat fill	entropy	$+0.19\sigma$	50% → **56%**
flat fill	probe	$+0.01\sigma$	50% → 50%

Table 19: Border perturbations touching no central pixel, 80 held-out images. Two structurally opposite perturbations both raise output entropy and neither moves the probe.

C An adversary on the allocation signal

The signal an attacker cannot see. A routing rule is an attack surface. Liu et al. [15] force deferral in multimodal cascades with a border-confined trigger that flattens the weak model's output distribution, inflating the provider's bill without changing any accuracy metric. Their attack is metric-agnostic precisely because it targets that distribution, which the probe never reads. As a cheap proxy we applied unoptimised border perturbations covering 36% of the image, with the probe fitted on training examples only (Table 19).

A perturbation requiring no optimisation, no model access and no knowledge of the deployed metric inflates compute by 24% relative when allocation reads output entropy, and by nothing when it reads the probe. We claim only the weak form: an attacker optimising the output distribution is not optimising a projection onto a direction they cannot observe, and here that indirection suffices. A white-box attacker who knows the probe direction is the case this does not settle.

D Abstention as a third action

We defined three targets in §3.3 and have used two. Q_3 (answerable at all?) is negative for 30.6% of the pilot: no configuration we run answers those queries correctly. A system that could recognise them would decline instead of spending, which is the third action the efficiency literature omits and the selective-prediction literature is built around [18, 19]. Two axes decide whether it is worth adding, and they disagree.

As a way to hold risk down, abstention is the weaker lever. Under a risk constraint the alternative to declining is to spend more and re-decide. Sweeping both thresholds and charging escalated queries for both passes (Table 20), escalation reaches coverage that abstention cannot.

risk tolerance	abstain only	or escalate	cost
$\leq 30\%$	unreachable	**58%**	320.8
$\leq 40\%$	41%	**88%**	312.2
$\leq 50\%$	69%	**100%**	200.7

Table 20: Maximum coverage at each risk tolerance, latency in milliseconds. At a 30% tolerance abstention alone serves no query at acceptable risk.

At a 50% tolerance escalation reaches full coverage while escalating 38% of traffic. The comparison needs a cost-instrumented multi-configuration run, which is the artefact the efficiency literature builds and the selective-prediction literature does not, and it says the two fields have been pulling different levers on the same signal.

As a way to save compute, abstention is nearly worthless here, because the rule got there first. Escalating a query no configuration can answer is pure waste, and 30.0% of the test fold is in that state. If the gain rule of §7 spent its escalation budget blind to solvability, a perfect abstention gate would recover that entire share. It does not (Table 21).

V	escalates	of which hopeless	perfect gate saves
400	24%	17%	2.8%
800	34%	19%	4.2%
1600	40%	22%	5.3%

Table 21: Escalation is aimed below the 30.0% base rate of unsolvable queries, so an oracle abstention gate (which cannot lose accuracy, since those queries are wrong under every action) recovers only the margin.

Calibrating on the signed gain already suppresses these queries: they have $\mathbb{E}[G \mid x] \approx 0$ whatever their error probability, so the rule declines them for the same reason it declines confident failures. The residual headroom for a perfect, free abstention detector is 4.2% of latency at $V = 800$. **A third action is not where the remaining budget is.** What the escalation budget actually buys is visible in the same run: 54% of escalated queries have zero gain, escalation changes nothing, and that, not unanswerability, is the waste worth attacking.

If a guarantee is needed rather than a preference. Both §7's rule and the thresholds above are calibrated to a point estimate. Where an operator needs a distribution-free promise instead, the three-way split of [13] transfers directly, with its middle regime spending compute rather than widening a prediction set: at answer-level $\alpha = 0.5$ the policy answers 52%, escalates 38% and declines 9%, and its coverage guarantee holds out of sample to within +0.025. A recall floor in the style of [12] gives the complementary guarantee (retain at least 90% of the queries escalation would have fixed, achieved 97% on the test fold) and cannot collapse to never escalating by construction, which is the failure mode our cost-minimising tuners exhibited. We report these as available machinery rather than as our headline policy, since neither improves on the cost-derived rule when a preference can be stated.

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